THE DIMENSION FORMULA FOR A SMOOTH MORPHISM OF LOCALLY NOETHERIAN ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement displays the dimension formula and names the relative dimension through a chapter definition; both are inside the counted statement span.

Editorial note. The three references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Dimension of an algebraic space at a point. Properties of Spaces, Definition spaces-properties-definition-dimension-at-point, used at proof lines 7549-7551. Gives the meaning of the dimensions in the formula.
- Relative dimension of a morphism at a point. Definition definition-dimension-fibre, used at proof lines 7544-7545. Fixes the term dim_x(X_y) in the statement.
- The scheme case for the dimension formula of a smooth morphism. Morphisms, Lemma morphisms-lemma-smoothness-dimension, used at proof lines 7552-7553. Is the input that closes the argument.

Source reference. Original definition 04NM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6085–6085. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 0AFF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13158–13158. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original definition 04N5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6066–6066. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let X and Y be locally Noetherian algebraic spaces over a scheme S, and let $f : X \to Y$ be a smooth morphism. For every point $x \in |X|$ with image $y \in |Y|$,*

$$\dim_x(X) = \dim_y(Y) + \dim_x(X_y)$$

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where $\dim_x(X_y)$ *is the relative dimension of* f *at* x *as in Definition 04NM.*

Proof. By definition of the dimension of an algebraic space at a point (Properties of Spaces, Definition 04N5), this reduces to the corresponding statement for schemes (Morphisms, Lemma 0AFF). □